

J3 Vector addition and multiplication

Name: _____

Class: _____

Date: _____

Time: **109 minutes**

Marks: **91 marks**

Comments:

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Q1.

Three forces act on a particle. These forces are $(9\mathbf{i} - 3\mathbf{j})$ newtons, $(5\mathbf{i} + 8\mathbf{j})$ newtons and $(-7\mathbf{i} + 3\mathbf{j})$ newtons. The vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors.

- (a) Find the resultant of these forces. (2)
- (b) Find the magnitude of the resultant force. (2)
- (c) Given that the particle has mass 5 kg, find the magnitude of the acceleration of the particle. (2)
- (d) Find the angle between the resultant force and the unit vector $\mathbf{i}$. (3)

(Total 9 marks)

Q2.

A particle moves with a constant acceleration of $(0.1\mathbf{i} - 0.2\mathbf{j}) \text{ m s}^{-2}$. It is initially at the origin where it has velocity $(-\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find an expression for the position vector of the particle t seconds after it has left the origin. (2)
- (b) Find the time that it takes for the particle to reach the point where it is due east of the origin. (3)
- (c) Find the speed of the particle when it is travelling south-east. (6)

(Total 11 marks)

Q3.

A helicopter is initially hovering above a lighthouse. It then sets off so that its acceleration is $(0.5\mathbf{i} + 0.375\mathbf{j}) \text{ m s}^{-2}$. The helicopter does not change its height above sea level as it moves. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find the speed of the helicopter 20 seconds after it leaves its position above the lighthouse. (4)
- (b) Find the bearing on which the helicopter is travelling, giving your answer to the nearest degree. (3)

- (c) The helicopter stops accelerating when it is 500 metres from its initial position.

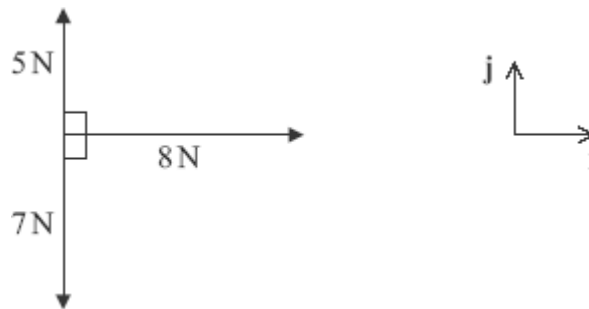
Find the time that it takes for the helicopter to travel from its initial position to the point where it stops accelerating.

(5)

(Total 12 marks)

Q4.

The diagram shows three forces and the perpendicular unit vectors $\mathbf{i}$ and $\mathbf{j}$, which all lie in the same plane.



- (a) Express the resultant of the three forces in terms of $\mathbf{i}$ and $\mathbf{j}$. (2)
- (b) Find the magnitude of the resultant force. (2)
- (c) Draw a diagram to show the direction of the resultant force, and find the angle that it makes with the unit vector $\mathbf{i}$. (3)

(Total 7 marks)

Q5.

A boat is initially at the origin, heading due east at 5 m s^{-1} . It then experiences a constant acceleration of $(-0.2\mathbf{i} + 0.25\mathbf{j}) \text{ m s}^{-2}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) State the initial velocity of the boat as a vector. (1)
- (b) Find an expression for the velocity of the boat t seconds after it has started to accelerate. (2)
- (c) Find the value of t when the boat is travelling due north. (3)
- (d) Find the bearing of the boat from the origin when the boat is travelling due north. (6)

(Total 12 marks)

Q6.

A particle P moves with acceleration $(-3\mathbf{i} + 12\mathbf{j}) \text{ m s}^{-2}$. Initially the velocity of P is $4\mathbf{i} \text{ m s}^{-1}$.

- (a) Find the velocity of P at time t seconds.

(2)

- (b) Find the speed of P when $t = 0.5$.

(3)

(Total 5 marks)

Q7.

Four forces, $\begin{bmatrix} 6 \\ 0 \end{bmatrix}$ newtons, $\begin{bmatrix} 0 \\ 4 \end{bmatrix}$ newtons, $\begin{bmatrix} -3 \\ -4.5 \end{bmatrix}$ newtons and $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$ newtons, act on a particle.

- (a) Express the resultant, $\mathbf{F}$ newtons, of these four forces as a column vector.

(2)

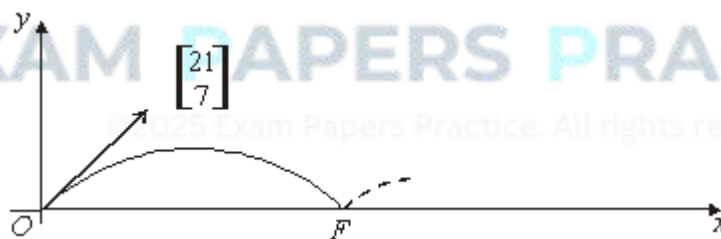
- (b) Find the magnitude of $\mathbf{F}$.

(2)

(Total 4 marks)

Q8.

A ball is kicked from a point O on horizontal ground with velocity $\begin{bmatrix} 21 \\ 7 \end{bmatrix} \text{ m s}^{-1}$. The ball subsequently hits the ground at a point F , as shown in the diagram.



- (a) (i) Find the time between the ball being kicked and reaching F .

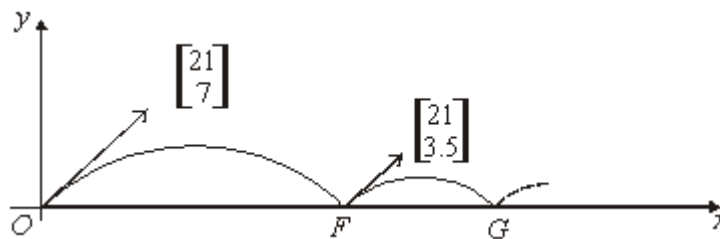
(3)

- (ii) Find the distance OF .

(2)

- (b) The ball rebounds from F with velocity $\begin{bmatrix} 21 \\ 3.5 \end{bmatrix} \text{ m s}^{-1}$. It subsequently hits the ground again at

G and then rebounds, as shown in the diagram below.



Find the distance FG .

(4)

- (c) After bouncing at G , the ball rebounds with velocity $\begin{bmatrix} 21 \\ 1.75 \end{bmatrix} \text{ m s}^{-1}$, so that the vertical component of the velocity is halved again, and then subsequently bounces next at H .

Find the total distance OH .

(3)

(Total 12 marks)

Q9.

A particle of mass 4 kg moves on a smooth horizontal plane. It is initially at rest at the origin. A force $\mathbf{F} = (8\mathbf{i} - 12\mathbf{j}) \text{ N}$ acts on the particle for 20 seconds. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the horizontal plane.

- (a) (i) Find the acceleration of the particle. (2)
- (ii) Find the velocity of the particle at the end of the 20 second period. (2)
- (iii) Find the position of the particle at the end of the 20 second period. (3)
- (b) At the end of the 20 second period, the force $\mathbf{F}$ is removed. Find the distance of the particle from the origin after the particle has been in motion for a total of 45 seconds. (5)

(Total 12 marks)

Q10.

Two forces $\mathbf{F}_1 = (3\mathbf{i} + 4\mathbf{j})\text{N}$ and $\mathbf{F}_2 = (6\mathbf{i} - 8\mathbf{j})\text{N}$, act on a particle. The resultant of these two forces is $\mathbf{F}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

- (a) Find $\mathbf{F}$. (2)
- (b) Find the magnitude of $\mathbf{F}$. (2)
- (c) Find the acute angle between $\mathbf{F}$ and the unit vector $\mathbf{i}$.

(3)
(Total 7 marks)



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